XRPL Validator — Daily Engineering Report · 2026-10-21

Period: 2026-10-21 00:00 to 24:00 US Eastern (2026-10-21 04:00 to 2026-10-22 04:00 UTC); ledgers #107,797,697 to #107,819,434. Every count from the ledger and the validator's logs stops at #107,819,434, closed 03:59:52 UTC. Generated 2026-10-22 00:20 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Status

Degraded — 1 engine receipt open.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 21,730 of the 21,738 ledgers the network closed (99.963%); every validation matched the network's validated ledger. Unsigned: 0 during planned restarts, 8 outside maintenance (1 incident, 1 isolated miss).
- **Native engine:** 2 engine receipts: #107,804,444 — fixed: commit e83a4f7 (2026-10-21); #107,806,242 — open.
- **Changes:** 0 deployments, 0 commits merged into main.
- **Network:** 4 slow-close periods; the longest, 13:17 UTC–13:22 UTC, closed ledgers up to 40 s apart with 6 empty ledgers.
- **Independence program:** 0 of 30 qualifying days (snapshot at 04:20:43 UTC).
- **Amendment fixCleanup3_4_0** holds majority; activation due ~2026-11-04 14:46 UTC.

Service levels

Measure	Value
Ledgers closed by the network	21,738
Validations issued	21,730
Coverage (validations / ledgers closed)	99.963%
Validations matching the network's validated ledger hash	21,730 of 21,730
Unsigned during planned restarts	0
Unsigned outside maintenance	8
Coverage outside planned maintenance (objective 99.9%)	99.963%
Restarts	0
Basis: our own ledger hash	21,499

Measure	Value
Basis: our own hash, witnessed by the reference node (quorum short)	231
External agreement, data.xrpl.org (UTC day 2026-10-21)	0.99534 (97 missed of 20,832)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#107,797,697–#107,819,434), by step:

Step	Ledgers with a receipt
Step 1 · native engine	0
Step 2 · records	1
Step 3 · order	1
Step 4 · peer data	1
Step 4 · validations	200
Step 5 · own hash	1
Writer	1

Live counters, snapshot at 04:20:43 UTC (since the last restart, 2026-10-21 02:03 UTC; not cut off at the period's last ledger):

Step	Checked	Result
1 · native engine vs libxrpl	23,688 ledgers, 4,100,320 transactions	23,688 fully matched; results 0, bytes 0, keys 0/0
2 · transaction records	23,766 ledgers, 4,210,830 records	4,210,822 exact; differ 8, tree 1
3 · transaction order	23,773 ledgers (mode: on)	23,772 exact; order 1, results 0, bytes 0, refused 1
4 · ledger data from peers	23,688 ledgers fetched	23,688 matched; differ 0, failed 0
4 · trusted validations	23,854 ledgers judged	23,639 agreed; differ 0, short of quorum 215
5 · our own ledger hash	23,856 ledgers	23,764 own; differ 1, partial 91
Writer	every ledger written	mismatch 0, distrusted 0

"Partial" ledgers are those the engine could not yet build alone while warming up after a restart.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	11:19 UTC	#107,804,444	Step 4 · peer data	the engine's inputs built from the peers' copy differed from the reference node's	fixed: commit e83a4f7 (2026-10-21); awaiting merge and deployment
R-02	13:17 UTC	#107,806,239–#107,806,257 (19)	Step 4 · validations	0–10 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (12 of 19 signed); see INC-01

ID	Time	Ledger(s)	Step	Description	Disposition
R-03	13:18 UTC	#107,806,242	Step 2 · records, Step 3 · order, Step 5 · own hash, Writer	OfferCreate tx 106994F46DD7: RippleState 2DE7C6A0 (modified on the network): FinalFields.Balance ours= {"currency":"524C55534400000000000…	open
R-04	14:07 UTC	#107,806,919–#107,806,934 (16)	Step 4 · validations	24–27 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)
R-05	15:08 UTC	#107,807,805–#107,807,834 (30)	Step 4 · validations	24–27 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)
R-06	15:12 UTC	#107,807,858–#107,807,902 (45)	Step 4 · validations	27 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)
R-07	15:52 UTC	#107,808,444–#107,808,448 (5)	Step 4 · validations	26 of the 27 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)
R-08	16:05 UTC	#107,808,638–#107,808,640 (3)	Step 4 · validations	27 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)
R-09	17:37 UTC	#107,810,010–#107,810,028 (19)	Step 4 · validations	26–27 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)
R-10	18:03 UTC	#107,810,398–#107,810,400 (3)	Step 4 · validations	27 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)
R-11	18:04 UTC	#107,810,413–#107,810,472 (60)	Step 4 · validations	27 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-10-21-01, INC-01 is INC-2026-10-21-01.

Incidents

INC-01 · 6 ledgers unsigned

Window	13:21:40 UTC to 13:22:50 UTC
Ledgers	#107,806,253–#107,806,258 (6)
Impact	the validator issued no validation for these ledgers; no incorrect validation was issued
Related receipts	R-02
Cause	Network-wide slow closes (13:17 UTC–13:22 UTC: up to 40 s apart, 6 empty ledgers). check_ledger returned no tree for a ledger with no transactions, so our own hash borrowed its transactions (and its order, which the native order never judges for an empty set). Under 5a, which signs only a ledger whose every part is ours, an empty ledger could never be signed. On mainnet the network resumed after the 21 Oct stall with six empty ledgers, #107806253-258, and the validator signed none of them, though our hash of each was the network's.
Action	commit 61be1f3 — fix(validator): an empty ledger's transaction tree is ours, so 5a can sign it; awaiting deployment

Isolated unsigned ledgers (shorter than 3 ledgers, cause not recorded): #107,806,242–#107,806,243 (2) at 13:18:40 UTC.

Change log

Deployments

No deployments.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Independence program

Qualifying days: **0 of 30** (snapshot at 04:20:43 UTC). A UTC day qualifies when every validation is of a ledger our own engine built, with our own transaction order and ledger data taken from peers, and coverage is at least 98%.

UTC day 2026-10-21 — tracked #107,794,029–#107,815,767	Value
Validations	21,633 (21,633 own, 0 borrowed)
Coverage of the tracked ledgers	99.51%
Our own transaction order	on
Ledger data from peers	no

Not qualifying: the ledger data came from the reference node, not peers (Step 4).

The validator saves this record every 64 validations, so a day in progress lags the signing log slightly; it is not cut off at the period's last ledger.

Network

Measure	Value
Ledgers closed	21,738 (#107,797,697–#107,819,434)
Mean close interval	3.97 s
Longest close interval	40 s (before #107,806,240)
Closes at least 15 s apart	14
Empty ledgers	6

- Slow closes 06:12:12 UTC–06:12:30 UTC: ledgers #107,799,728, 1 close at least 15 s apart, up to 18 s.
- Slow closes 13:17:20 UTC–13:22:40 UTC: ledgers #107,806,240–#107,806,257 (18), 11 closes at least 15 s apart, up to 40 s; empty: #107,806,253, #107,806,254, #107,806,255, #107,806,256, #107,806,257, #107,806,258.
- Slow closes 14:07:10 UTC–14:07:30 UTC: ledgers #107,806,920, 1 close at least 15 s apart, up to 20 s.
- Slow closes 14:08:00 UTC–14:08:20 UTC: ledgers #107,806,925, 1 close at least 15 s apart, up to 20 s.

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Voting events:

- fixCleanup3_4_0 lost majority — ledger 107,805,697, 2026-10-21 12:41 UTC
- fixCleanup3_4_0 gained majority — ledger 107,807,489, 2026-10-21 14:46 UTC

Amendment	Majority since	Activation due
fixCleanup3_4_0	2026-10-21 14:46 UTC	2026-11-04 14:46 UTC

Amendments enabled on mainnet: 97.

Open items

- Receipt R-03 (#107,806,242): open, under investigation.
- Receipt R-01 (#107,804,444): fixed: commit e83a4f7 (2026-10-21); awaiting merge and deployment.
- Unsigned #107,806,242–#107,806,243 (2) (13:18:40 UTC): cause not recorded.
- Committed, awaiting merge and deployment: `61be1f3` fix(validator): an empty ledger's transaction tree is ours, so 5a can sign it.
- Amendment fixCleanup3_4_0: activation due ~2026-11-04 14:46 UTC.

Corrections to earlier reports

None.